

RCC Driver Analysis

The RCC (Reset and Clock Control) driver is responsible for providing clock information required by different peripherals in the STM32F446xx microcontroller. Every peripheral works based on a clock signal, so knowing the peripheral clock frequency is important for configuring communication interfaces like USART, SPI, and I2C.

In this project, the RCC driver is implemented using two files:

- **stm32f446xx_rcc_driver.h** – Contains function declarations.
- **stm32f446xx_rcc_driver.c** – Contains the implementation of clock calculation functions.

The driver provides APIs to calculate the APB1 and APB2 peripheral clock frequencies.

How the RCC Driver Works

The RCC driver reads the clock configuration from the RCC configuration register (CFGR). Based on the selected clock source and prescaler values, it calculates the clock frequency supplied to different buses.

The driver first checks whether the system clock is generated from:

- HSI (High Speed Internal oscillator – 16 MHz)
- HSE (High Speed External oscillator – 8 MHz)

After identifying the clock source, it reads the AHB and APB prescaler values from the RCC register. Using these values, it calculates the final peripheral clock frequency.

The calculated clock value is returned to the application and can be used for configuring communication peripherals

How the Driver Hides Hardware Complexity

Without the RCC driver, the programmer would have to read multiple bits from the RCC configuration register and manually calculate the clock frequency.

The RCC driver performs all these calculations internally and provides simple APIs that return the required clock values.

Instead of accessing registers directly, the application only needs to call:

```
RCC_GetPCLK1Value();
```

```
RCC_GetPCLK2Value();
```

This makes the program simpler and avoids calculation errors.

Prescaler Arrays

The driver uses lookup tables to convert prescaler bits into actual division factors.

AHB Prescaler

```
uint16_t AHB_PreScaler[8] = {2,4,8,16,64,128,256,512};
```

These values are used when the AHB prescaler is enabled.

APB Prescaler

```
uint8_t APB1_PreScaler[4] = {2,4,8,16};
```

These values are used to determine the APB1 and APB2 clock division factors.

APIs Provided by the Driver

1. RCC_GetPCLK1Value()

This function calculates and returns the APB1 peripheral clock frequency.

Working

- Reads the system clock source from RCC->CFGR.
- Determines whether HSI or HSE is selected.
- Reads the AHB prescaler value.
- Reads the APB1 prescaler value.
- Calculates the final APB1 clock using:

$$\text{PCLK1} = \text{System Clock} / \text{AHB Prescaler} / \text{APB1 Prescaler}$$

The returned value is mainly used for configuring peripherals connected to the APB1 bus such as USART2, USART3, SPI2, SPI3, and I2C peripherals.

2. RCC_GetPCLK2Value()

This function calculates and returns the APB2 peripheral clock frequency.

Working

- Reads the system clock source.
- Reads the AHB prescaler value.
- Reads the APB2 prescaler value.

- Calculates the final APB2 clock.

PCLK2 = System Clock / AHB Prescaler / APB2 Prescaler

The returned clock value is used by peripherals connected to the APB2 bus such as USART1, USART6, SPI1, SPI4, ADC, and SYSCFG.

Advantages of the RCC Driver

- Simplifies clock frequency calculation.
- Eliminates manual register calculations.
- Improves code readability.
- Makes peripheral configuration easier.
- Can be reused in different embedded projects.
- Provides accurate clock values required for communication drivers.

The RCC driver provides a simple method to obtain APB1 and APB2 peripheral clock frequencies without directly accessing RCC registers. By internally reading the clock source and prescaler settings, it performs all necessary calculations and returns the correct clock value to the application. This driver reduces programming complexity and helps other drivers such as USART, SPI, and I2C configure their baud rates and communication timings correctly.